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HONORS CHEMISTRY • UNIT 5

Moles and Molar Quantities

Count particles by the mole, then connect particles, grams and gas volume through molar mass.

TEACHER EDITION

Everything in a light-gray box, cell or line is an answer or model response. Teacher notes (misconceptions, interpretation choices, corrections) appear in the heavy-bordered boxes labeled TEACHER NOTE. The pages follow the student packet, so they line up closely.

TEACHER OVERVIEW

Numbering: the homework folder calls this "Unit 5 Topic Homework" (Topics 5.1–5.6), the same unit number as this packet (between Chemical Reactions, Unit 4, and Stoichiometry, Unit 6). The packet has topics 5.1–5.7: homework 5.1 is split into 5.1 (the mole) and 5.2 (molar mass); homework 5.2 = 5.3, 5.3 = 5.4, 5.4 = 5.5, 5.5 = 5.6, 5.6 = 5.7.

Answers are computed. Every numerical answer in the key was calculated by code from the formulas, using atomic masses to 0.01 g/mol (H 1.01, C 12.01, N 14.01, O 16.00, Na 22.99, Mg 24.31, Al 26.98, P 30.97, S 32.06, Cl 35.45, K 39.10, Ca 40.08, Fe 55.85, Cu 63.55, Ag 107.87), $N_(A_) = 6.022 \times 10^{23}$ and 22.4 L/mol at STP. Unrounded values carry through; only the final answer is rounded to the sig figs of the data. If your class uses different masses the last digit may differ.

Homework quirks handled: 5.1 #8 writes lead(II) carbonate as $PbCO_4$, which is not a real formula; the packet uses $PbCO_3$. 5.2 #6 uses Cu_2O_3 (copper(III) oxide), unusual but kept, because the copper count does not depend on the oxide. 5.4 #3 uses N_2S_2 (normally written S_2N_2); kept as given. Numbers such as "3.4x10²⁶" are written in scientific notation, and units, state of the gas (STP) and sig figs were added where the homework leaves them out. Each topic has a method box, a worked example and 1–2 new multiple-choice items, and an Extra Practice page of new problems was written.

Overlap with other units: Unit 1 (topic 1.4) first introduces the mole, Avogadro's number and molar mass; this unit goes deeper and adds STP volume, percent composition, formulas and hydrates. The Stoichiometry unit uses these conversions (including STP volume in 7.3) as tools.

Common misconceptions: a mole is a mass (it is a number); 1 mol of every substance has the same mass; using the atomic number as the mass; forgetting the subscript outside parentheses; multiplying when you should divide (and the reverse); using 22.4 L/mol away from STP or for a liquid or solid; rounding a ratio such as 1.5 to 2 instead of doubling; reading the water of a hydrate as part of the salt's molar mass.

Pacing idea: 5.1–5.2 on day 1; 5.3 (mole map until students can draw it) on day 2; 5.4 on day 3; 5.5–5.6 on days 4–5; 5.7 with the hydrate lab and Extra Practice on day 6.

What we will learn

	Topic
5.1	The mole and Avogadro's number
5.2	Molar mass
5.3	Mole conversions (particles, moles and grams)
5.4	Volume of a gas at STP
5.5	Percent composition

	Topic
5.6	Empirical and molecular formulas
5.7	Hydrates

By the end of this unit I can...

- Explain what a mole is and use Avogadro's number to convert between particles and moles.
- Find the molar mass of an element or compound, including formulas with parentheses.
- Convert between particles, moles and grams using the mole map, showing every unit.
- Use 22.4 L/mol to convert between moles, grams, particles and the volume of a gas at STP.
- Calculate the percent composition by mass of a compound.
- Find an empirical formula from masses or percentages, and a molecular formula from the molar mass.
- Use the mass lost on heating to find the formula and name of a hydrate.

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5.1 The Mole and Avogadro's Number

Counting by the mole • 6.022×10^{23} • atoms, molecules and formula units

BIG IDEA

Atoms and molecules are far too small to count one at a time, so chemists count them in huge groups called **moles**. 1 mole = **6.022×10^{23}** particles, just as 1 dozen = 12. A mole is a number, not a mass.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Mole	*What is it, and what does it measure?* The SI unit for amount of substance: 6.022×10^{23} particles of anything.	1 mol of eggs = 6.022×10^{23} eggs
Avogadro's number	*Write the value in scientific notation. What does it count?* 6.022×10^{23} particles per mole, the number of particles in 1 mole.	1 mol = 6.022×10^{23} particles
Particle	*Name the three kinds of particles and say which substances each goes with.* Atoms (single elements), molecules (covalent compounds and diatomic elements) and formula units (ionic compounds).	Al → atoms; H_2O → molecules; NaCl → formula units

HOW TO CONVERT BETWEEN PARTICLES AND MOLES

1. Decide what the particles are: atoms, molecules or formula units.
2. Write **1 mol = 6.022×10^{23} particles** as a fraction so the unit you start with cancels.
3. Moles to particles: **multiply** by 6.022×10^{23}. Particles to moles: **divide** by 6.022×10^{23}.
4. Round to the sig figs of the data and write the unit.

WORKED EXAMPLE

How many atoms are in 2.50 mol of magnesium?

Work it out here:

2.50 mol Mg × (6.022×10^{23} atoms / 1 mol) = **1.51×10^{24} atoms Mg**

1. What is a "mole" in chemistry, and what is it used to measure?

A mole is a counting unit, 6.022×10^{23} particles. It measures the amount of a substance (how many particles there are) the way a dozen counts eggs.

2. What is the difference between an atom and a molecule?

An atom is the smallest unit of an element. A molecule is two or more atoms bonded together (H_2O has 3 atoms; O_2 has 2).

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Convert (show the fraction setup)

	Particles and moles
3.	How many atoms are in 1.75 moles of aluminum? 1.75 mol × (6.022 × 10^{23} / 1 mol) = **1.05 × 10^{24} atoms Al**
4.	How many molecules are in 3.25 moles of H_2O? 3.25 mol × (6.022 × 10^{23} / 1 mol) = **1.96 × 10^{24} molecules H_2O**
5.	How many moles are in 3.4 × 10^{26} molecules of CO_2? 3.4 × 10^{26} molecules × (1 mol / 6.022 × 10^{23}) = **5.6 × 10^{2} mol CO_2**
6.	How many moles are there in 3.88 × 10^{25} atoms of nickel? 3.88 × 10^{25} atoms × (1 mol / 6.022 × 10^{23}) = **64.4 mol Ni**

TEACHER NOTE

Students often treat the mole as a mass. Hold up "a dozen" and "a mole" of different objects: the number is the same, the mass is not. #3 and #6 count atoms (elements), #4 and #5 count molecules (covalent compounds). #5 is limited to 2 sig figs by 3.4 × 10^{26}, so 5.6 × 10^{2} mol (565 is acceptable with the unit).

1. Which sample contains the most particles?

A) 1 mol of iron B) 1 mol of water C) 1 mol of helium D) They all contain the same number

Answer: D — 1 mol of anything is 6.022 × 10^{23} particles.

2. How many oxygen atoms are in 1 mol of O_2 molecules?

A) 6.022 × 10^{23} B) 1.204 × 10^{24} C) 3.011 × 10^{23} D) 32

Answer: B — 2 atoms per molecule × 6.022 × 10^{23}

5.2 Molar Mass *Grams per mole • add the atomic masses • subscripts and parentheses*

BIG IDEA

The **molar mass** of a substance is the mass of 1 mole of it, in **g/mol**. For a compound, add the atomic masses of every atom in the formula. Molar mass is the bridge between grams and moles.

HOW TO FIND A MOLAR MASS

1. List each element in the formula and **count** its atoms. A subscript outside parentheses multiplies everything inside: in $Mg(OH)_2$ there are 2 O and 2 H.
2. Multiply each count by the atomic mass (to 0.01 g/mol).
3. Add the results. Round to 0.01 g/mol and write the unit **g/mol**.

WORKED EXAMPLE

Find the molar mass of aluminum sulfate, $Al_2(SO_4)_3$.

Work it out here:
Al: 2 × 26.98 = 53.96 S: 3 × 32.06 = 96.18 O: 12 × 16.00 = 192.00
Molar mass = 53.96 + 96.18 + 192.00 = **342.14 g/mol**

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Molar mass	*What does it measure, and what unit does it have?* The mass of one mole of a substance, in grams per mole (g/mol).	H_2O: 18.02 g/mol
Diatomic element	*Name the seven that exist as pairs of atoms.* An element found as two bonded atoms: H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2.	O_2: 2 × 16.00 = 32.00 g/mol

1. Define molar mass in your own words.

The mass in grams of one mole (6.022×10^{23} particles) of a substance; its unit is g/mol.

Calculate the molar mass of each substance (show the sum)

	Formula	Work and molar mass (g/mol)
1.	Be	**9.01 g/mol**
2.	N_2	**2 × 14.01 = 28.02 g/mol**
3.	O_2	**2 × 16.00 = 32.00 g/mol**
4.	F_2	**2 × 19.00 = 38.00 g/mol**
5.	Mg	**24.31 g/mol**
6.	Ag	**107.87 g/mol**
7.	LiI	**Li 6.94 + I 126.90 = 133.84 g/mol**
8.	$PbCO_3$	**Pb 207.20 + C 12.01 + O 3(16.00) = 267.21 g/mol**
9.	$Mg(OH)_2$	**Mg 24.31 + O 2(16.00) + H 2(1.01) = 58.33 g/mol**
10.	K_3PO_4	**K 3(39.10) + P 30.97 + O 4(16.00) = 212.27 g/mol**

	Formula	Work and molar mass (g/mol)
11.	$(NH_4)_3PO_4$	**N 3(14.01) + H 12(1.01) + P 30.97 + O 4(16.00) = 149.12 g/mol**
12.	$C_6H_{12}O_6$	**C 6(12.01) + H 12(1.01) + O 6(16.00) = 180.18 g/mol**

TEACHER NOTE — CORRECTIONS

The homework lists #8 as $PbCO_4$. That formula does not exist (it would need Pb^{6+}); lead(II) carbonate is $PbCO_3$, so the key uses $PbCO_3$ = 267.21 g/mol. Check that students count the atoms inside parentheses: $Mg(OH)_2$ is 24.31 + 2(16.00) + 2(1.01) = 58.33, not 24.31 + 16.00 + 1.01. Diatomic elements (N_2, O_2, F_2) are the usual slip: 14.01 is the atomic mass of nitrogen, 28.02 is the molar mass of N_2.

LAB CORNER • Weighing a mole

A student weighs out exactly 1 mole of each of three substances: 18.02 g of water, 58.44 g of table salt (NaCl) and 55.85 g of iron.

1. (a) Which sample contains the most particles? (b) Why are the three masses different?

Answer:

(a) None: each one is 1 mole, so each has 6.022×10^{23} particles.

(b) The particles have different masses. A mole of heavy particles has a greater mass than a mole of light ones.

2. What is the molar mass of $Ca(OH)_2$?

A) 57.09 g/mol B) 74.10 g/mol C) 58.10 g/mol D) 73.09 g/mol

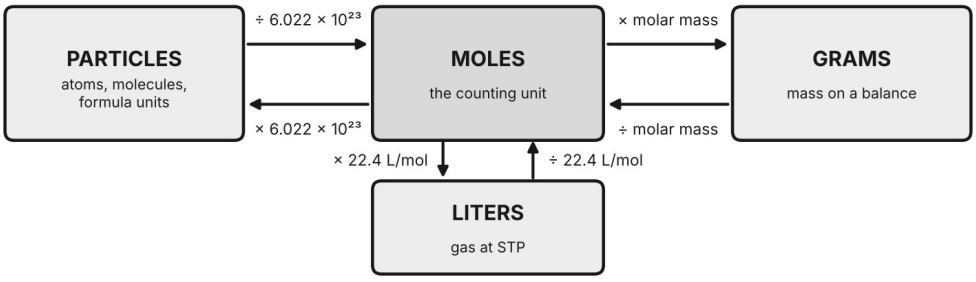
Answer: B — 40.08 + 2(16.00) + 2(1.01) = 74.10

5.3 Mole Conversions *Particles • moles • grams • the mole map*

BIG IDEA

You can never go straight from particles to grams. **Every conversion goes through moles.** The mole map shows which factor to use for each arrow: Avogadro's number for particles, molar mass for grams.

Every conversion goes through moles



The mole map with every conversion factor filled in.

STEPS

1. Write the given quantity with its unit.
2. Find the path on the mole map. Grams or particles go to **moles first**, then on to the answer.
3. Set up each factor as a fraction so the unit you want to remove cancels.
4. Calculate, round to the sig figs of the data, and label the answer with the unit and the substance.

WORKED EXAMPLE

How many molecules are in 9.00 g of water?

Work it out here:

molar mass of H_2O = 18.02 g/mol

9.00 g × (1 mol / 18.02 g) × (6.022 × 10^{23} molecules / 1 mol) = **3.01 × 10^{23} molecules H_2O**

Solve each problem. Show givens, factors and the answer with units.

	Mole conversion problems
1.	Calculate the number of grams of CO_2 that will contain 2.5 × 10^{24} molecules of CO_2. 2.5 × 10^{24} molecules × (1 mol / 6.022 × 10^{23}) × (44.01 g / 1 mol) = **1.8 × 10^2 g CO_2**
2.	How many moles of carbon tetrafluoride, CF_4, are in 176 grams of CF_4? molar mass CF_4 = 12.01 + 4(19.00) = 88.01 g/mol; 176 g × (1 mol / 88.01 g) = **2.00 mol CF_4**
3.	How many moles of water are in 360.0 grams of water? 360.0 g × (1 mol / 18.02 g) = **19.98 mol H_2O**
4.	How many moles of iron(III) oxide, Fe_2O_3, are there in 79.85 grams of iron(III) oxide? molar mass Fe_2O_3 = 159.70 g/mol; 79.85 g × (1 mol / 159.70 g) = **0.5000 mol Fe_2O_3**
5.	How many molecules of chlorine gas, Cl_2, are present in 52.5 grams of Cl_2? 52.5 g × (1 mol / 70.90 g) × (6.022 × 10^{23} / 1 mol) = **4.46 × 10^{23} molecules Cl_2**
6.	How many copper atoms are present in 5.6 moles of Cu_2O_3? 5.6 mol Cu_2O_3 × (2 mol Cu / 1 mol Cu_2O_3) × (6.022 × 10^{23} atoms / 1 mol) = **6.7 × 10^{24} atoms Cu**
7.	A student measures out 2.5 moles of glucose, $C_6H_{12}O_6$. How many grams of oxygen are contained in that amount of glucose? 2.5 mol $C_6H_{12}O_6$ × (6 mol O / 1 mol $C_6H_{12}O_6$) × (16.00 g O / 1 mol O) = **2.4 × 10^2 g O**

	Mole conversion problems
8.	A student measures out 15.0 grams of $CaCl_2$. How many grams of chlorine are present? 15.0 g × (1 mol $CaCl_2$ / 110.98 g) × (2 mol Cl / 1 mol $CaCl_2$) × (35.45 g Cl / 1 mol Cl) = **9.58 g Cl**

TEACHER NOTE

#6, #7 and #8 find the amount of **one element inside a compound**: the subscript is a mole ratio (2 mol Cu per mol Cu_2O_3, 6 mol O per mol glucose, 2 mol Cl per mol $CaCl_2$). #6 gives the same answer for Cu_2O, CuO or Cu_2O_3 because only the 2 Cu matters. #1 and #7 are limited to 2 sig figs. Common error: dividing by Avogadro's number when going from moles to particles.

1. To convert grams of CO_2 to molecules of CO_2, the first step is to…

A) multiply by 6.022×10^{23} B) divide by the molar mass C) multiply by 22.4 D) divide by 6.022×10^{23}

Answer: B — Grams go to moles first.

2. What is the mass of 0.50 mol of H_2O?

A) 9.0 g B) 18 g C) 36 g D) 0.50 g

Answer: A — 0.50 × 18.02 = 9.0 g

5.4 Volume of a Gas at STP *22.4 L/mol • 0 °C and 1 atm • moles, grams and particles*

BIG IDEA

At **STP** (standard temperature and pressure: **0 °C and 1 atm**), 1 mole of **any** gas takes up **22.4 L**. That adds one more arrow to the mole map: moles ↔ liters of gas.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
STP	*What do the letters stand for, and what are the two conditions?* Standard temperature and pressure: 0 °C (273 K) and 1 atm.	A balloon at 0 °C and 1 atm
Molar volume	*How much space does 1 mol of gas fill at STP? Does the kind of gas matter?* 22.4 L per mole at STP, the same for every gas.	1 mol He = 1 mol CO_2 = 22.4 L

STEPS

1. Moles ↔ liters: use **1 mol = 22.4 L** (STP only, gases only).
2. If the given is grams or particles, go to **moles first** with the molar mass or Avogadro's number.
3. Then use 22.4 L/mol, or go on to grams or particles for the answer.

WORKED EXAMPLE

What volume does 8.00 g of O_2 occupy at STP?

Work it out here:
8.00 g × (1 mol / 32.00 g) × (22.4 L / 1 mol) = **5.60 L O_2**

What volume will each of the following occupy at STP?

	Volume at STP
1.	3.20 moles of O_2 3.20 mol × (22.4 L / 1 mol) = **71.7 L**
2.	1.75 moles of NH_3 1.75 mol × 22.4 L/mol = **39.2 L**
3.	37.8 moles of CO_2 37.8 mol × 22.4 L/mol = **847 L**
4.	5.05 grams of helium 5.05 g × (1 mol / 4.00 g) × (22.4 L / 1 mol) = **28.3 L**
5.	60.0 grams of xenon 60.0 g × (1 mol / 131.29 g) × (22.4 L / 1 mol) = **10.2 L**

Solve each problem. All gases are at STP.

	Gas problems at STP
6.	A sample of NH_3 gas occupies 75.3 L at STP. How many molecules is this? 75.3 L × (1 mol / 22.4 L) × (6.022×10^{23} / 1 mol) = **2.02×10^{24} molecules NH_3**
7.	What is the mass of 575 L of ammonia (NH_3) gas at STP? 575 L × (1 mol / 22.4 L) × (17.04 g / 1 mol) = **437 g NH_3**
8.	A 5.00 gram sample of O_2 is in a container at STP. What is the volume of this sample? 5.00 g × (1 mol / 32.00 g) × (22.4 L / 1 mol) = **3.50 L O_2**
9.	How many molecules are present in the container in the question above? 5.00 g × (1 mol / 32.00 g) × (6.022×10^{23} / 1 mol) = **9.41×10^{22} molecules O_2**
10.	How many molecules are present in 0.12 L of carbon dioxide gas at STP? How many grams? 0.12 L × (1 mol / 22.4 L) = 0.0054 mol CO_2 molecules: × 6.022×10^{23} = **3.2×10^{21} molecules** grams: × 44.01 g/mol = **0.24 g CO_2**

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TEACHER NOTE

22.4 L/mol is exact enough for the class; the molar volume at STP is 22.414 L/mol. Remind students that 22.4 L/mol is for **gases at STP only**, not for liquids, solids or other temperatures: those use the ideal gas law in the Gases unit. #9 reuses the mole value from #8, so students should carry it forward rather than re-round. #10 asks for both molecules and grams from the same mole value.

1. Which has the greatest volume at STP?

A) 1.00 mol of N_2 B) 28.02 g of N_2 C) 6.022×10^{23} molecules of N_2 D) All three are the same

Answer: D — Each is 1.00 mol of N_2 = 22.4 L.

2. Why can't you use 22.4 L/mol for a gas at 25 °C?

A) 22.4 L/mol applies only at 0 °C and 1 atm B) The gas has no molar mass at 25 °C C) Moles change with temperature D) 22.4 L/mol is only for noble gases

Answer: A

5.5 Percent Composition *Part ÷ whole × 100 • mass percent of each element*

BIG IDEA

Percent composition tells you what part of a compound's mass comes from each element. The percents of all the elements add up to **100 %**.

HOW TO FIND PERCENT COMPOSITION

1. Find the molar mass of the compound (the **whole**).
2. For each element, find its mass in 1 mole of the compound (atomic mass × subscript): the **part**.
3. **% element = (mass of element in 1 mol ÷ molar mass of compound) × 100.**
4. Check that the percents add to 100 (within rounding).

WORKED EXAMPLE

Find the percent composition of calcium carbonate, $CaCO_3$.

Work it out here:

molar mass = 40.08 + 12.01 + 3(16.00) = 100.09 g/mol

Ca: 40.08 / 100.09 × 100 = 40.04% C: 12.01 / 100.09 × 100 = 12.00% O: 48.00 / 100.09 × 100 = 47.96%

Check: 40.04 + 12.00 + 47.96 = 100.00 %

Find the percent composition of every element in each compound

	Percent composition
1.	NaOH molar mass = Na 22.99 + O 16.00 + H 1.01 = 40.00 g/mol **Na 57.48% O 40.00% H 2.53%**
2.	$CuBr_2$ molar mass = Cu 63.55 + Br 2(79.90) = 223.35 g/mol **Cu 28.45% Br 71.55%**

	Percent composition
3.	N_2S_2 molar mass = N 2(14.01) + S 2(32.06) = 92.14 g/mol **N 30.41% S 69.59%**
4.	$(NH_4)_2S$ molar mass = N 2(14.01) + H 8(1.01) + S 32.06 = 68.16 g/mol **N 41.11% H 11.85% S 47.04%**
5.	$(NH_4)_3PO_4$ molar mass = N 3(14.01) + H 12(1.01) + P 30.97 + O 4(16.00) = 149.12 g/mol **N 28.19% H 8.13% P 20.77% O 42.92%**

TEACHER NOTE

Percents are rounded to 0.01 %, so a total can be 99.99 % or 100.01 % (NaOH: 57.48 + 40.00 + 2.53 = 100.01). That is a rounding effect, not a mistake. N_2S_2 is unusual (the real compound is written S_2N_2) but the calculation is the same. For $(NH_4)_3PO_4$, count 3 N, 12 H, 1 P and 4 O. Common error: using the mass of one atom instead of the mass of all the atoms of that element in the formula.

1. A 50.0 g sample of a compound is 20.0 % hydrogen by mass. How many grams of hydrogen are in the sample?

A) 2.5 g B) 10.0 g C) 20.0 g D) 40.0 g

Answer: B — 0.200 × 50.0 g = 10.0 g

2. Which has the greater percent oxygen by mass?

A) H_2O B) H_2O_2 C) They are the same D) Cannot be told

Answer: B — H_2O_2: 32.00/34.02 = 94.06 %; H_2O: 16.00/18.02 = 88.79 %

5.6 Empirical and Molecular Formulas

Simplest whole-number ratio • percent → grams → moles → ratio

BIG IDEA

The **empirical formula** is the simplest whole-number ratio of atoms in a compound. The **molecular formula** is the actual number of each atom, and it is a whole-number multiple of the empirical formula: **molecular = n × empirical**, where **n = molar mass ÷ empirical formula mass**.

STEPS

1. Change each percent to grams (assume a **100 g** sample), or use the grams given.
2. Change each mass to **moles** (÷ atomic mass).
3. Divide every mole value by the **smallest**. These are the subscripts.
4. If a result is near x.5, multiply **all** by 2 (near x.33 or x.67, by 3). Do not round 1.5 up to 2.
5. Molecular formula: n = molar mass ÷ empirical mass; multiply every subscript by n.

WORKED EXAMPLE

A compound is 40.0 % C, 6.7 % H and 53.3 % O and has a molar mass of 180 g/mol. Find its empirical and molecular formulas.

Work it out here:

100 g → 40.0 g C, 6.7 g H, 53.3 g O. moles: C 40.0/12.01 = 3.33, H 6.7/1.01 = 6.6, O 53.3/16.00 = 3.33

÷ smallest (3.33): C 1.0, H 2.0, O 1.0 → empirical formula **CH_2O**

empirical mass = 12.01 + 2(1.01) + 16.00 = 30.03; n = 180 / 30.03 = 6.0 ≈ 6 → molecular formula **$C_6H_{12}O_6$**

Solve each problem

	Empirical and molecular formula problems
1.	Determine the empirical formula of the compound if it contains 87.3 g Na, 121.5 g S and 91.2 g O. moles: Na 3.80, S 3.79, O 5.70. ÷ 3.79: 1 : 1 : 1.5 → ×2 → 2 : 2 : 3 empirical formula = **$Na_2S_2O_3$**
2.	A compound contains 85.65 % Bi, 6.56 % O and 7.79 % F. What is the empirical formula of this compound? moles: Bi 0.4098, O 0.410, F 0.410 → ratio 1 : 1 : 1 empirical formula = **BiOF**
3.	Propene has an empirical formula of CH_2. What is its molecular formula if it has a molar mass of 42.09 g/mol? empirical mass = 12.01 + 2(1.01) = 14.03; n = 42.09 / 14.03 = 3.00 ≈ 3 molecular formula = **C_3H_6**
4.	The percent composition of an unknown substance is 75.42 % carbon, 6.63 % hydrogen, 8.38 % nitrogen and 9.57 % oxygen. If its molar mass is 334.0 g/mol, what is its empirical and molecular formula? moles: C 6.280, H 6.56, N 0.598, O 0.598. ÷ 0.598: 10.5 : 11.0 : 1 : 1 → ×2 → 21 : 22 : 2 : 2 empirical formula = **$C_{21}H_{22}N_2O_2$** (mass 334.45 g/mol); n = 334.0 / 334.45 = 0.999 ≈ 1 molecular formula = **$C_{21}H_{22}N_2O_2$**

TEACHER NOTE

#1: the ratio is 1 : 1 : 1.5, so double everything ($Na_2S_2O_3$, sodium thiosulfate). #4 is the hard one: the ratio is 10.5 : 10.97 : 1 : 1 (the hydrogen value is 11 within the rounding of the percents), so double to 21 : 22 : 2 : 2. The empirical mass (334.45 g/mol) matches the given molar mass, so the molecular formula equals the empirical formula (n = 1). Common errors: rounding 1.5 to 2, dividing by the largest value, and skipping the step that changes percent to grams.

1. What is the empirical formula of $C_6H_{12}O_6$?

A) CHO B) CH_2O C) $C_2H_4O_2$ D) $C_6H_{12}O_6$

Answer: B — Divide every subscript by 6.

2. A compound has the empirical formula CH and a molar mass of 78 g/mol. Its molecular formula is…

A) CH B) C_2H_2 C) C_6H_6 D) C_6H_{12}

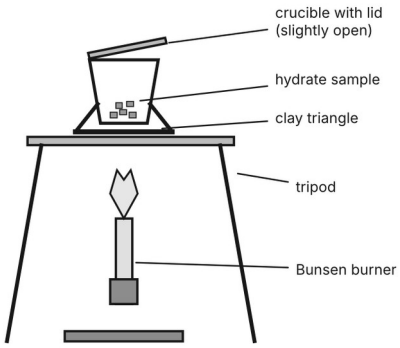
Answer: C — 78 / 13.02 = 6

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5.7 Hydrates *Water in the crystal • heating • the formula and the name*

BIG IDEA

A **hydrate** is an ionic compound with a fixed number of water molecules built into its crystal, such as $CuSO_4 \cdot 5H_2O$. Heating drives the water off and leaves the **anhydrous** ("without water") salt. The mass lost is the mass of water, so you can compare moles of water to moles of salt.



Heating a hydrate in a crucible. Mass the crucible, the hydrate and the anhydrous salt.

1	2	3	4	5	6	7	8	9	10
mono-	di-	tri-	tetra-	penta-	hexa-	hepta-	octa-	nona-	deca-

HOW TO FIND THE FORMULA OF A HYDRATE

1. mass of water = mass of hydrate − mass of anhydrous salt.
2. Change both to **moles**: water ÷ 18.02 g/mol, salt ÷ its molar mass.
3. x = moles of H_2O ÷ moles of salt (divide by the smaller). Round to the **nearest whole number**.
4. Write the formula **salt · xH_2O** and the name **salt name + prefix + "hydrate"**.

WORKED EXAMPLE

A 7.36 g sample of a hydrate of calcium chloride is heated until no water remains. The $CaCl_2$ left has a mass of 5.55 g. Find the formula and name of the hydrate.

Work it out here:

water = 7.36 − 5.55 = 1.81 g; mol H_2O = 1.81 / 18.02 = 0.100 mol

mol $CaCl_2$ = 5.55 / 110.98 = 0.0500 mol; x = 0.100 / 0.0500 = 2.01 ≈ 2

formula **$CaCl_2 \cdot 2H_2O$**; name **calcium chloride dihydrate**

Find the formula and the name of each hydrate

	Hydrate problems
1.	A hydrate of magnesium sulfate has a mass of 13.52 g. This sample is heated until no water remains. The $MgSO_4$ anhydrate has a mass of 6.60 g. Find the formula and name of the hydrate. water = mass lost = 6.92 g → 0.384 mol H_2O; $MgSO_4$: 0.0548 mol; x = 7.00 ≈ 7 formula **$MgSO_4 \cdot 7H_2O$**; name **magnesium sulfate heptahydrate**

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	Hydrate problems
2.	A sample of copper(II) sulfate hydrate has a mass of 3.97 g. After heating, the $CuSO_4$ that remains has a mass of 2.54 g. Determine the correct formula and name of the hydrate. water = mass lost = 1.43 g → 0.0794 mol H_2O; $CuSO_4$: 0.0159 mol; x = 4.99 ≈ 5 formula **$CuSO_4 \cdot 5H_2O$**; name **copper(II) sulfate pentahydrate**
3.	A sample of the hydrate of sodium carbonate has a mass of 8.85 g. It loses 1.28 g when heated. Find the formula and the name of the hydrate. water = mass lost = 1.28 g → 0.0710 mol H_2O; Na_2CO_3: 0.0714 mol; x = 0.995 ≈ 1 formula **$Na_2CO_3 \cdot H_2O$**; name **sodium carbonate monohydrate**
4.	A 16.4 g sample of hydrated calcium sulfate is heated until all the water is driven off. The calcium sulfate that remains has a mass of 13.0 g. Find the formula and the chemical name of the hydrate. water = mass lost = 3.40 g → 0.189 mol H_2O; $CaSO_4$: 0.0955 mol; x = 1.98 ≈ 2 formula **$CaSO_4 \cdot 2H_2O$**; name **calcium sulfate dihydrate**
5.	A hydrate is determined to be 45.43 % water and 54.57 % $CoCl_2$. Find the chemical formula and name for this hydrate. (Hint: assume that there is 100 g total of hydrate compound.) 100 g → 45.43 g H_2O = 2.521 mol; 54.57 g $CoCl_2$ ÷ 129.83 = 0.4203 mol; x = 6.00 ≈ 6 formula **$CoCl_2 \cdot 6H_2O$**; name **cobalt(II) chloride hexahydrate**

TEACHER NOTE

#3 gives x = 0.99, which rounds to 1 (a monohydrate; the prefix "mono-" is used). #4 is gypsum. Students should name the anhydrous salt first (use the nomenclature rules, including the Roman numeral for copper and cobalt) and then add the prefix. A hydrate lab fits here: heat blue $CuSO_4 \cdot 5H_2O$ gently until it turns white, with the lid slightly open and goggles on, and let the sample cool before it is massed. Common error: dividing the mass of water by the mass of the salt instead of comparing **moles**.

1. After a hydrate is heated, its mass decreases. The mass lost is the mass of…

A) the salt B) the water C) the crucible D) air

Answer: B

2. Which is the correct name for $CoCl_2 \cdot 6H_2O$?

A) cobalt chloride hexahydrate B) cobalt(II) chloride hexahydrate C) cobalt(II) chloride sixhydrate D) cobalt(II) chlorate hexahydrate

Answer: B

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PRACTICE

Extra Practice — Unit 5

Show your work. Every conversion goes through moles.

A. Which conversion? (give the factor you would use)

	Situation	Factor
a.	Grams of a compound to moles of the compound.	÷ molar mass (g/mol)
b.	Moles of molecules to the number of molecules.	× 6.022×10^{23} molecules/mol
c.	Liters of a gas at STP to moles of the gas.	÷ 22.4 L/mol
d.	Percent composition to grams (to start an empirical formula).	assume 100 g: each % becomes grams

B. Calculate

	Problems
a.	How many molecules are in 8.50 g of NH_3? 8.50 g × (1 mol / 17.04 g) × (6.022×10^{23} / 1 mol) = **3.00×10^{23} molecules NH_3**
b.	What is the percent by mass of carbon in glucose, $C_6H_{12}O_6$? 6(12.01) / 180.18 × 100 = **39.99%**
c.	A compound is 52.14 % C, 13.13 % H and 34.73 % O and has a molar mass of 46.07 g/mol. Find its molecular formula. moles: C 4.34, H 13.0, O 2.17 → ratio 2 : 6 : 1; empirical mass 46.07, n = 1 → **C_2H_6O**
d.	What is the mass of 5.00 L of CO_2 at STP? 5.00 L × (1 mol / 22.4 L) × (44.01 g / 1 mol) = **9.82 g CO_2**

C. Lab data: a hydrate of barium chloride

A student heats a hydrate of barium chloride in a crucible. Complete the table, then find x in $BaCl_2 \cdot xH_2O$ (molar mass of $BaCl_2$ = 208.23 g/mol).

Measurement	Data	Your calculation
Mass of empty crucible	18.42 g	

Measurement	Data	Your calculation
Mass of crucible + hydrate	21.67 g	
Mass of crucible + anhydrous $BaCl_2$ after heating	21.19 g	
Mass of the hydrate		3.25 g
Mass of the anhydrous $BaCl_2$		2.77 g
Mass of water lost		0.48 g

1. Find the moles of water, the moles of $BaCl_2$, the value of x, and write the formula and name.

Answer:
water: 0.48 g / 18.02 = 0.0266 mol; $BaCl_2$: 2.77 g / 208.23 = 0.0133 mol; x = 2.00 ≈ 2
formula **$BaCl_2 \cdot 2H_2O$**; name **barium chloride dihydrate**

D. Think

1. One mole of copper (63.55 g) and one mole of helium (4.00 g) have different masses. Explain why they still contain the same number of atoms.

A mole is a number of particles (6.022×10^{23}), not a mass. Copper atoms have a much larger mass than helium atoms, so a mole of copper has more mass.

2. A student finds the formula of a hydrate to be $CuSO_4 \cdot 4.2H_2O$. Give one possible reason the ratio is not a whole number, and say what the student should report.

The sample may not have been heated long enough (some water left), or the masses were read wrongly. The ratio should be a whole number, so the student reports $CuSO_4 \cdot 4H_2O$ only after checking, and ideally reheats and reweighs.

3. Which of these is NOT 1 mole of the substance?

A) 6.022×10^{23} atoms of Na B) 22.99 g of Na C) 22.4 L of Na at STP D) 58.44 g of NaCl

Answer: C — Sodium is a solid; 22.4 L/mol is only for gases at STP.

4. A compound has the empirical formula CH_2O and a molar mass of 60.06 g/mol. Its molecular formula is…

A) CH_2O B) $C_2H_4O_2$ C) $C_3H_6O_3$ D) $C_6H_{12}O_6$

Answer: B — 60.06 / 30.03 = 2